Kabzozo Supermarket UK – Data Analysis Report

1. Executive Summary

This report presents a structured analysis of supermarket transaction data covering 5,053 records across 2024. The analysis applies a rigorous workflow including data validation, transformation, statistical testing, and visual exploration to derive actionable business insights.

The most critical findings are: revenue is concentrated in a few product categories, customer membership is strongly associated with higher spending, and sales exhibit clear seasonal variation with a mid-year decline and strong year-end recovery.

2. Dataset Overview

The dataset consists of 5,053 transaction records with 9 variables including customer attributes, product categories, pricing, quantity, payment method, and ratings. Data quality is high with no missing values or duplicates.

check	value
rows	5053
columns	12
invalid_dates	0
total_missing_cells	0
duplicate_rows	0
nonpositive_unit_price	0

3. Analytical Framework

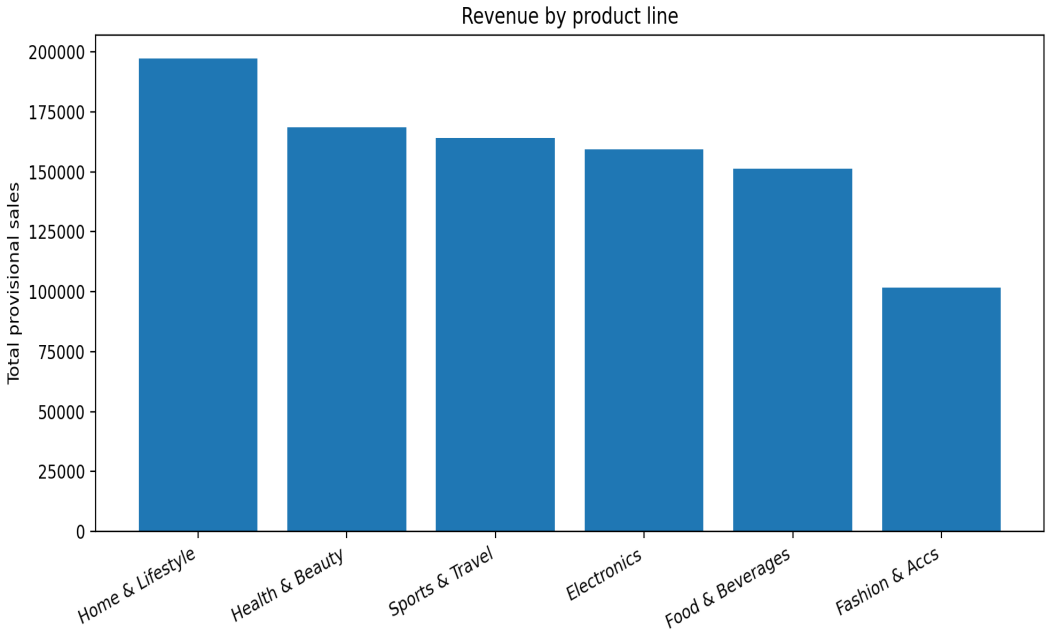
The analysis follows a structured workflow: Data Description → Inquiry → Goal Design → Statistical Analysis → Visualization. This ensures transparency, reproducibility, and traceability across all outputs.

4. Detailed Findings

4.1 Revenue by Product Line

ct line	Transactions	Units	Total_Sales	Avg_Sales	Median_Sales	Avg_Rating	Revenue_Share
& Lifestyle	708	3461	197260.56	278.61661016949154	247.44	6.779378531073447	20.9285064790
n & Beauty	958	4296	168570.61	175.96097077244258	125.64	6.867327766179541	17.8846248006
s & Travel	576	2697	164252.94	285.1613541666667	224.025	6.728993055555556	17.4265383763
onics	465	2325	159259.1	342.492688172043	270.99	6.82258064516129	16.8967131908

& Beverages	1670	7309	151357.73	90.63337125748504	31.09	7.084191616766467	16.0584114379
on & Accs	676	3123	101843.91	150.65667159763314	85.525	6.697189349112426	10.8052057151

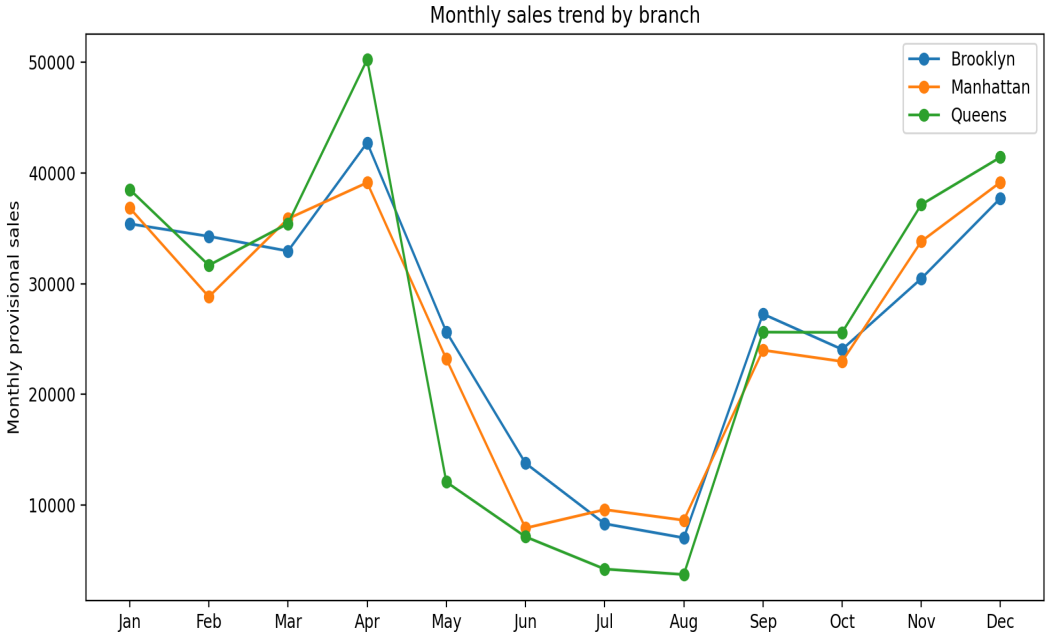


Revenue is concentrated in a few categories, with Home & Lifestyle leading.

What this means in simple terms… A few categories drive most of the business.

4.2 Branch Performance

Branch	Transactions	Total_Sales	Avg_Sales	Median_Sales	Avg_Rating	Revenue_Share_%
Brooklyn	1793	319705.48	178.30757389849413	103.6	6.87133296151701	33.919391740350605
Queens	1555	312842.82	201.18509324758844	115.89	6.871189710610933	33.19129270081949
Manhattan	1705	309996.55	181.8161583577713	109.08	6.909090909090909	32.8893155588299

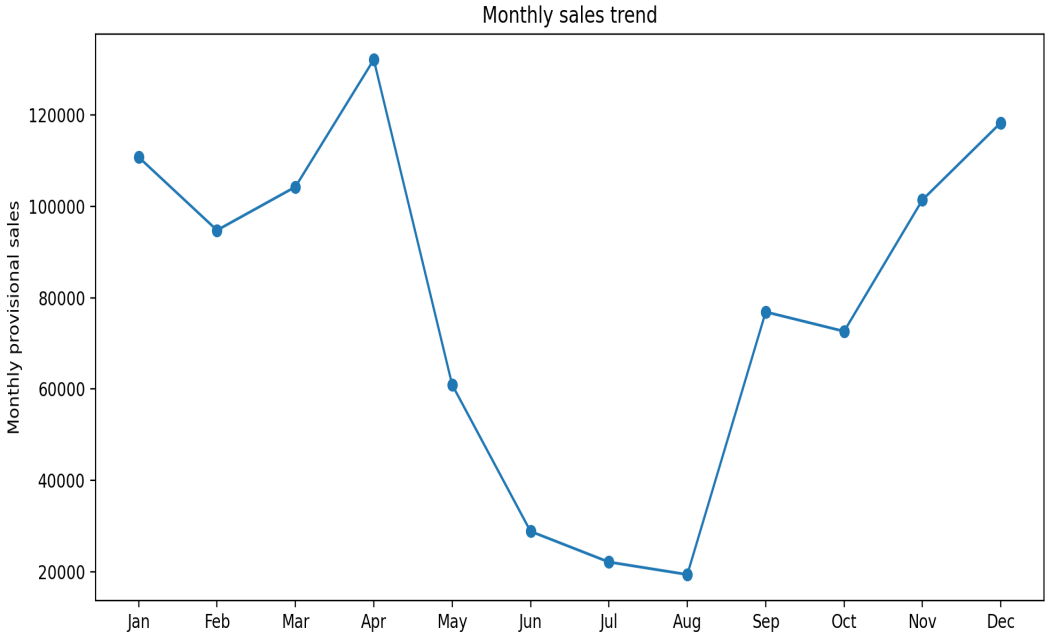


Branches have similar totals but differ in transaction value efficiency.

What this means in simple terms… Some branches earn more per sale, not just in total.

4.3 Monthly Sales Trends

Month	Transactions	Total_Sales	Avg_Sales	MoM_Growth_%	Month_Name
1	352	110754.16	314.6425	nan	Jan
2	311	94761.64	304.69980707395496	-14.439656262121447	Feb
3	345	104243.34	302.15460869565214	10.00584202637269	Mar
4	447	132083.88	295.4896644295302	26.707260147266965	Apr
5	417	60955.78	146.1769304556355	-53.85070456743094	May
6	416	28883.41	69.43127403846154	-52.61579787839643	Jun



Sales show strong seasonal patterns with mid-year decline.

What this means in simple terms… Sales are not consistent throughout the year.

4.4 Customer Type Spending

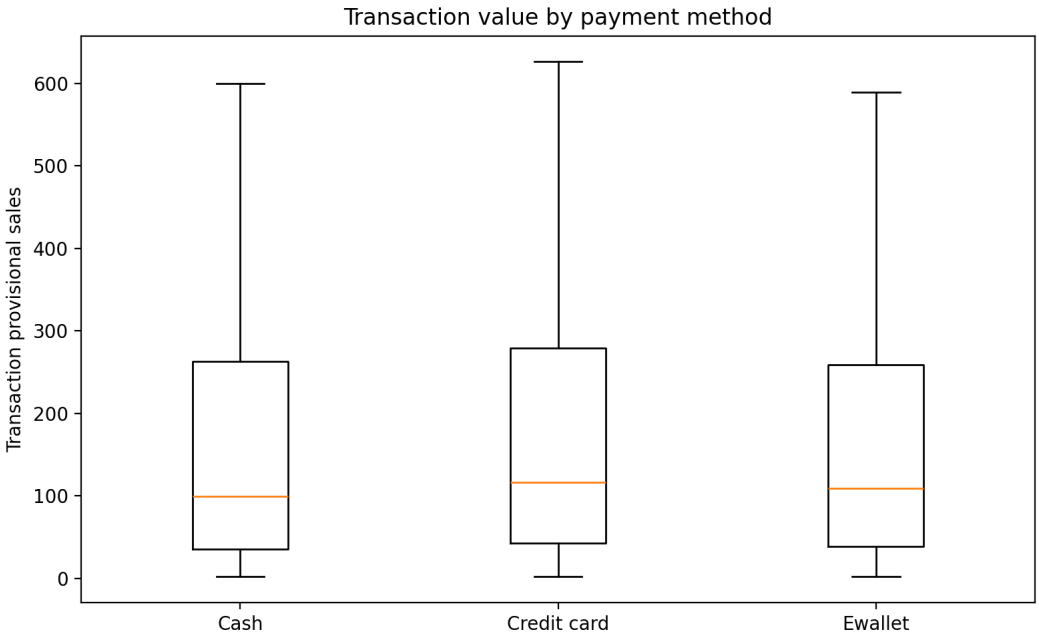
Customer type	Transactions	Total_Sales	Avg_Sales	Median_Sales	Avg_Rating
Member	2085	436368.17	209.2892901678657	133.15	6.872230215827338
Normal	2968	506176.68	170.54470350404313	96.58	6.892318059299192

Members spend more per transaction than non-members.

What this means in simple terms… Loyalty customers are more valuable.

4.5 Payment Analysis

Payment	Transactions	Total_Sales	Avg_Sales	Median_Sales
Cash	1674	304622.65	181.9729091995221	99.0
Credit card	1594	311230.89	195.25149937264743	116.325
Ewallet	1785	326691.31	183.0203417366947	108.74

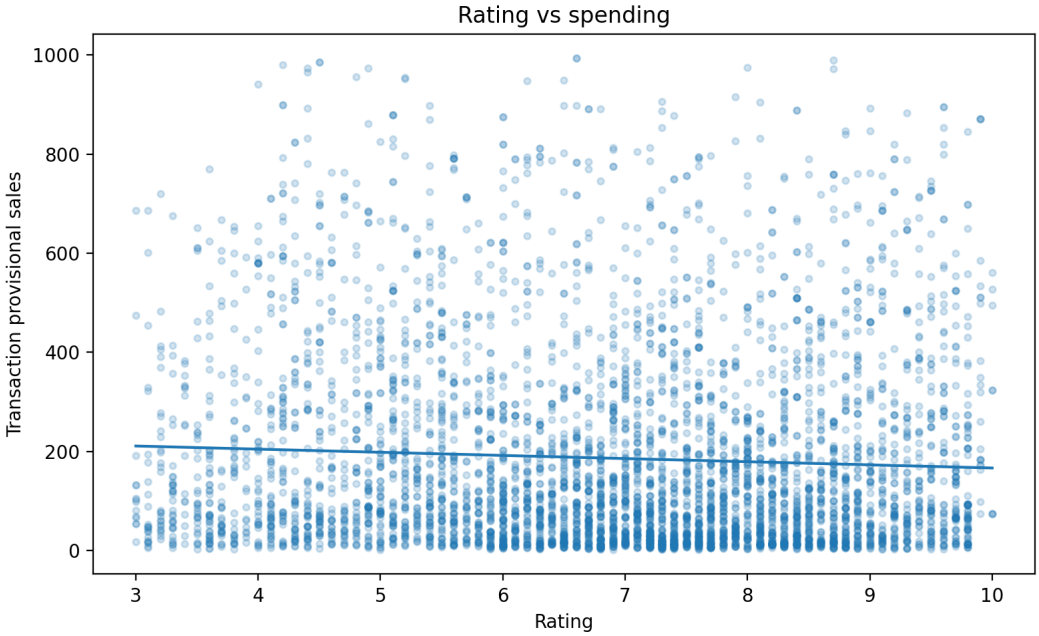


Payment method influences transaction size slightly.

What this means in simple terms… Card users tend to spend more.

4.6 Rating vs Spending

metric	value
Pearson_r	-0.0541069403948047
Pearson_p	0.0001190716232432
Spearman_rho	-0.0624478431708273
Spearman_p	8.899874073706806e-06
OLS_slope	-6.33361461200408
OLS_intercept	230.13252212122737

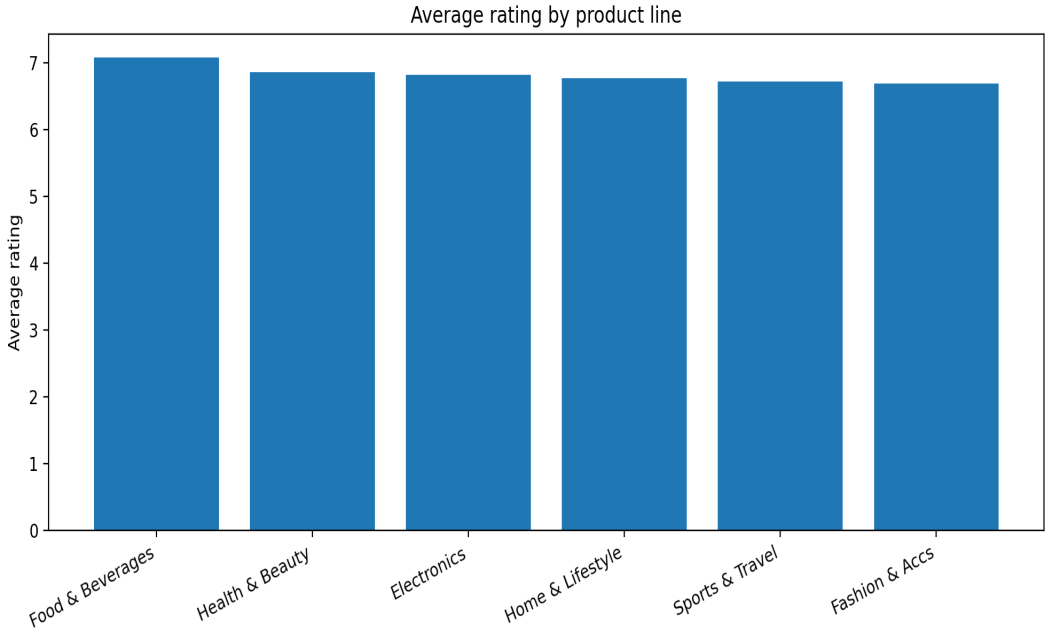


Correlation between rating and spending is weak.

What this means in simple terms… Spending does not strongly affect satisfaction.

4.7 Product Ratings

Product line	Transactions	Avg_Rating	Median_Rating	Rating_SD
Food & Beverages	1670	7.084191616766467	7.3	1.6415264671671326
Health & Beauty	958	6.867327766179541	7.0	1.6818122779386242
Electronics	465	6.82258064516129	6.8	1.8308732272316264
Home & Lifestyle	708	6.779378531073447	7.0	1.7043329947330972
Sports & Travel	576	6.728993055555556	6.8	1.7770350567715782
Fashion & Accs	676	6.697189349112426	6.8	1.7357609750804734

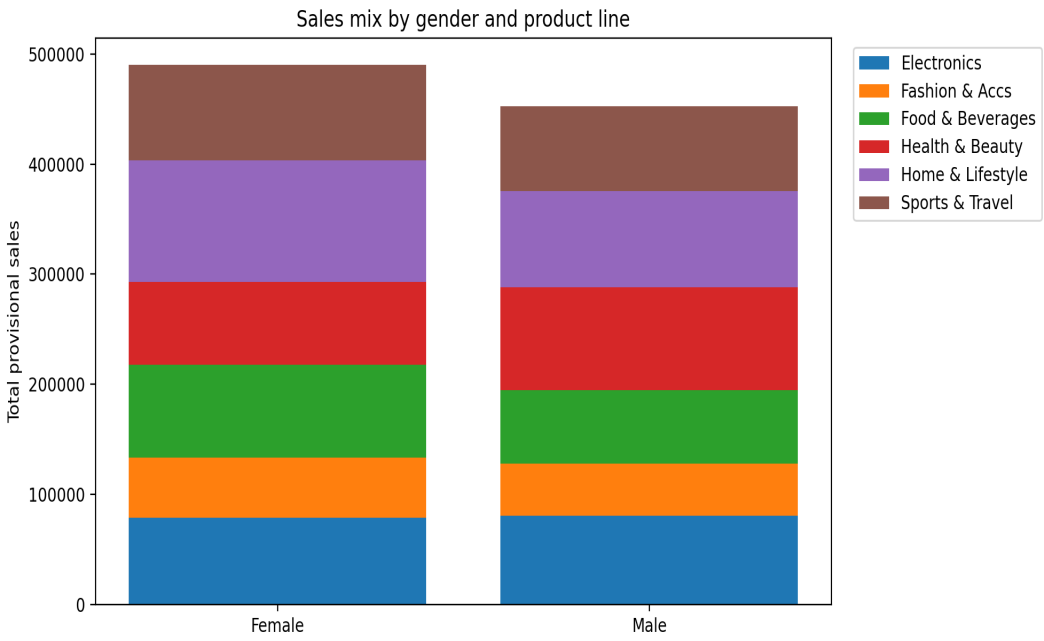


Customer satisfaction varies by product category.

What this means in simple terms… Some products perform better in customer experience.

4.8 Gender Analysis

Gender	Transactions	Total_Sales	Avg_Sales	Median_Sales	Avg_Rating
Female	2603	490169.72	188.309535151748	111.82	6.884402612370342
Male	2450	452375.13	184.64291020408163	106.84	6.883632653061225

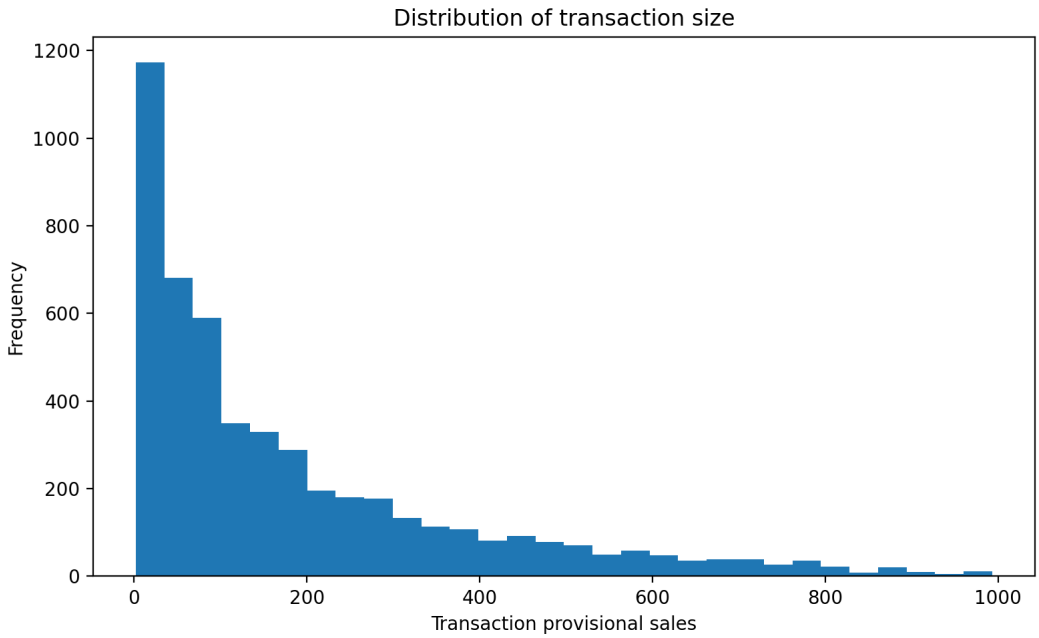


Some differences exist in product preferences by gender.

What this means in simple terms… Preferences vary slightly between groups.

4.9 Transaction Distribution

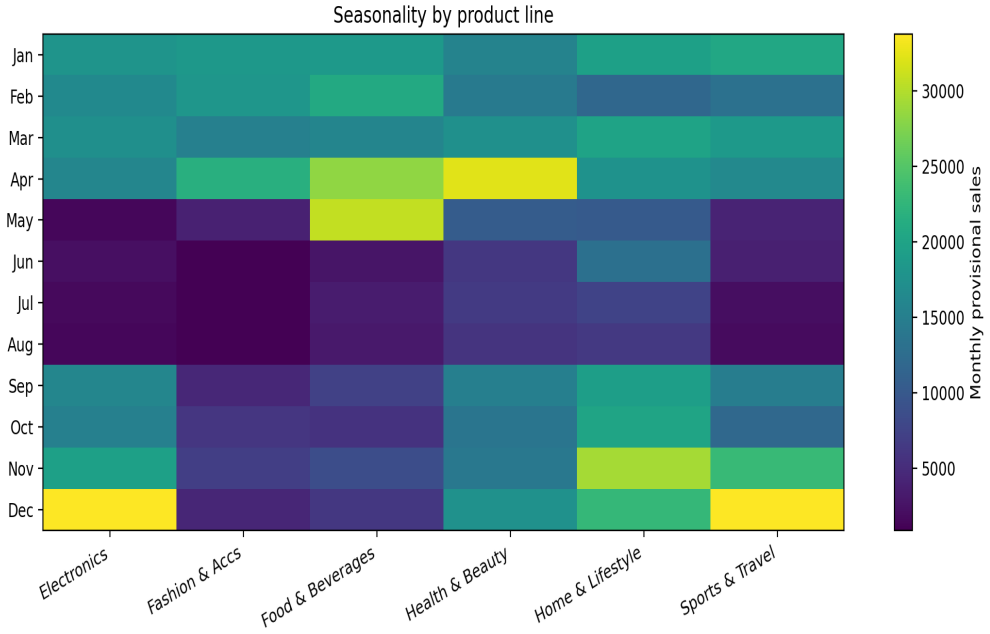
metric	value
mean	186.53173362358996
median	109.6
std	200.1966967646809
min	2.18
max	993.0
skewness	1.5568503889139835



Transaction values are skewed with few high-value purchases.

What this means in simple terms… A few large purchases raise the average.

4.10 Seasonality by Product Line



Seasonal effects vary across product lines.

What this means in simple terms… Different products peak at different times.

5. Cross-Domain Insights

Revenue concentration, customer segmentation, and seasonality collectively shape business performance. High-value categories combined with loyal customer segments present the strongest growth opportunities.

6. Strategic Recommendations

1. Prioritise high-performing product categories for inventory and promotions.

2. Strengthen loyalty programs to increase customer lifetime value.

3. Align operations with seasonal demand patterns.

7. Visual Appendix

All charts and tables are included within the document and accompanying data files.

8. Methodological Note

The analysis follows a structured and reproducible workflow ensuring data integrity, analytical transparency, and consistent interpretation across all findings.